

61. Put $x^{\frac{1}{3}} = t$
62. $y - 2 = \frac{1}{4 + y - 2} = \frac{1}{y + 2} \Rightarrow y = \pm\sqrt{5} (y > 0)$
63. $ax^2 + bx + c = 0$
 $(a+k)x^2 + (b+k)x + (c+k) = 0$
 $\therefore \frac{a+k}{a} = \frac{b+k}{b} = \frac{c+k}{c}$
 $\frac{k}{a} = \frac{k}{b} = \frac{k}{c} \therefore a = b = c \Rightarrow x^2 + x + 1 = 0$
64. Let $f(x) = ax^2 + bx + c$ equation whose roots $\frac{1-\alpha}{\alpha}, \frac{1-\beta}{\beta}$ is $f\left(\frac{1}{x+1}\right) = 0 \therefore px^2 + qx + r = 0$
 & $c(x+1)^2 + b(x+1) + a = 0$ are same.
65. $\frac{1}{3} < \frac{x^2 + 2x + 4}{x^2 - 2x + 4} < 3 \forall x \in R$
 $\frac{93 \cdot 2^x + 6 \cdot 3^x + 4}{93 \cdot 2^x - 6 \cdot 3^x + 4} = \frac{(3^{x+1})^2 + 2 \cdot 3^{x+1} + 4}{(3^{x+1})^2 - 2 \cdot 3^{x+1} + 4}$
 $= \frac{y^2 + 2y + 4}{y^2 - 2y + 4}$ when $3^{x+1} = y$
66. min value $= \frac{4ac - b^2}{4a} = \frac{1}{1+b^2} \therefore m(b) = \frac{1}{1+b^2}$
 Range $= (0, 1]$
67. $\frac{(x-1)(4x-3) - (x-3)(4x+5)}{(4x+5)(4x-3)} < 0$
 $\Rightarrow \frac{18}{(4x+5)(4x-3)} < 0$
 $\therefore (4x+5)(4x-3) < 0; \therefore x \in \left(-\frac{5}{4}, \frac{3}{4}\right)$
68. We have, $(\log_5 x)^2 + \log_5 x < 2$
 Put $\log_5 x = a$, then
 $a^2 + a < 2 \Rightarrow (a+2)(a-1) < 0$
 $\Rightarrow -2 < a < 1$ or $-2 < \log_5 x < 1$
 $\therefore 5^{-2} < x < 5$ or $\frac{1}{25} < x < 5$

69. $\therefore x \in (-\infty, -5) \cup (1, 2)$
 From the diagram of wave nature of the curve
70. solve $x(x^3 + ax + 1) = 0$ i.e.,
 $x^4 + ax^2 + x = 0$ and $x^4 + ax^2 + 1 = 0$
71. range $= \left(f\left(\sqrt{\frac{c}{a}}\right), f\left(-\sqrt{\frac{c}{a}}\right) \right)$
72. $\underline{R} = \frac{\pi}{4} \therefore P + Q = \frac{3\pi}{4}$
 $\tan\left(\frac{P}{3} + \frac{Q}{3}\right) = \tan\left(\frac{\pi}{4}\right) \Rightarrow a + b = c$
73. $\alpha + \beta = \frac{b}{5 + \sqrt{2}}$ and $\alpha\beta = \frac{8 + 2\sqrt{5}}{5 + \sqrt{2}}$ given that
 harmonic mean between the roots of the given equation is 4.
 $\frac{2\alpha\beta}{\alpha + \beta} = 4; b = \frac{8 + 2\sqrt{5}}{2} = 4 + \sqrt{5}$



LEVEL-II (H.W)

1. If α, β are the roots of the equation $8x^2 - 3x + 27 = 0$, then the value of $\left(\frac{\alpha^2}{\beta}\right)^{\frac{1}{3}} + \left(\frac{\beta^2}{\alpha}\right)^{\frac{1}{3}}$ is
 1) $\frac{1}{3}$ 2) $\frac{1}{4}$ 3) $\frac{7}{2}$ 4) 4
2. If α, β are the roots of $x^2 - px + q = 0$ then the equation whose roots are $\alpha\beta + \alpha + \beta$, $\alpha\beta - \alpha - \beta$ is
 1) $x^2 + 2qx + p^2 + q^2 = 0$
 2) $x^2 - 2qx + p^2 + q^2 = 0$
 3) $x^2 - 2qx + q^2 - p^2 = 0$
 4) $x^2 + 2qx + q^2 - p^2 = 0$
3. If $\sin \theta, \cos \theta$ are the roots of the equation $ax^2 + bx + c = 0$ then
 1) $a^2 - b^2 + 2ac = 0$ 2) $a^2 + b^2 + 2ac = 0$
 3) $a - b + 2ac = 0$ 4) $a + b + 2c = 0$

4. If α, β are the roots of $x^2 - x + 1 = 0$ then $\alpha^5 + \beta^5 =$
 1) 2 2) 1 3) 12 4) 4
5. If α, β are the roots of $3x^2 + 5x - 7 = 0$ then the value of $\left(\frac{1}{3\alpha+5}\right)^2 + \left(\frac{1}{3\beta+5}\right)^2$ is
 1) $\frac{67}{63}$ 2) $\frac{67}{441}$ 3) $\frac{109}{441}$ 4) $\frac{109}{63}$
6. If the sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to sum of their squares, then
 1) $ab + b^2 + 2ac = 0$ 2) $ab + a^2 + 2ac = 0$
 3) $ab + b^2 - 2ac = 0$ 4) $ab + a^2 - 2ac = 0$
7. If α, β are the roots of $6x^2 - 6x + 1 = 0$ then $\frac{1}{2}(a + ba + c\alpha^2 + d\alpha^3) + \frac{1}{2}(a + b\beta + c\beta^2 + d\beta^3) =$
 1) $a + b + c + d$
 2) $a + 2b + 3c + 4d$
 3) $a + (b/2) + (c/3) + (d/4)$ 4) 0
8. If α, β are the roots of $x^2 + ax - b = 0$ and γ, δ are the roots of $x^2 + ax + b = 0$ then $(\alpha - \gamma)(\alpha - \delta)(\beta - \delta)(\beta - \gamma) =$
 1) $4b^2$ 2) b^2 3) $2b^2$ 4) $3b^2$
9. If the arithmetic mean of the roots of a quadratic equation is $8/5$ and the arithmetic mean of their reciprocal is $8/7$ then the equation is
 1) $5x^2 + 16x + 7 = 0$ 2) $5x^2 - 16x + 7 = 0$
 3) $7x^2 + 16x + 5 = 0$ 4) $7x^2 - 16x + 5 = 0$
10. If one root of the equation $8x^2 - 6x + k = 0$ is the square of the other, then $k =$
 1) 0,3 2) -1,27 3) 0,-2 4) 1,-27
11. If the ratio of the roots of $ax^2 + 2bx + c = 0$ is same as the ratio of the roots of $px^2 + 2qx + r = 0$ then
 1) $\frac{b^2}{ac} = \frac{p^2}{qr}$ 2) $\frac{b}{ac} = \frac{q}{pr}$
 3) $\frac{b^2}{ac} = \frac{q^2}{pr}$ 4) $\frac{b}{ac} = \frac{q^2}{pr}$

12. If the difference of the roots of the equation $x^2 - bx + c = 0$ is equal to the difference of the roots of the equation $x^2 - cx + b = 0$ and $b \neq c$ then $b + c =$
 1) 0 2) 2 3) 4 4) -4
13. Which one of the following Means of the roots of the equation $x^2 - 2bx + a^2 = 0$ is the A.M. of the roots of $x^2 - 2ax + b^2 = 0$?
 1) A.M 2) G.M 3) H.M 4) A.G.P
14. If $\tan A, \tan B$ are the roots of $x^2 - 2x + 2 = 0$ then $\sin^2(A + B) =$
 1) $4/5$ 2) $1/2$ 3) $3/5$ 4) $1/4$
15. If the roots of $\frac{x^2 - bx}{ax - c} = \frac{k - 1}{k + 1}$ are numerically equal but opposite in sign then $k =$
 1) c 2) $\frac{1}{c}$ 3) $\frac{a+b}{a-b}$ 4) $\frac{a-b}{a+b}$

NATURE OF THE ROOTS AND PROPERTIES

16. If $pr = 2(q + s)$ then among the equations $x^2 + px + q = 0$ and $x^2 + rx + s = 0$
 1) both have real roots
 2) both have imaginary roots
 3) at least one has real roots
 4) at least one has imaginary roots.
17. The roots of the equation $(b - c)x^2 + 2(c - a)x + (a - b) = 0$ are always
 1) real and distinct 2) real and equal
 3) real 4) imaginary
18. The roots of the equation $(a + c - b)x^2 + 2cx + (b + c - a) = 0$ ($a \neq b$) are
 1) real and distinct 2) real and equal
 3) real 4) imaginary
19. If the roots of the equation $x^2 - 2cx + ab = 0$ be real and unequal, the roots of the equation $x^2 - 2(a + b)x + (a^2 + b^2 + 2c^2) = 0$ are
 1) real and distinct 2) real and equal
 3) real 4) imaginary

20. If a, b, c, d are real and no two of them are simultaneously zero, then the equation

$$(x^2 + ax - 3b)(x^2 - cx + b) \\ (x^2 - dx + 2b) = 0 \quad \text{has}$$

- 1) at least two real roots
- 2) at least four real roots
- 3) all roots real
- 4) at least two imaginary roots

21. a, b, c are rational. Then the roots of

$$(a + b + c)x^2 - 2(a + c)x + (a - b + c) = 0 \quad \text{are}$$

- 1) real
- 2) equal
- 3) rational
- 4) imaginary

SOLVING EQUATIONS

22. If the equation

$$(\lambda^2 - 5\lambda + 6)x^2 + (\lambda^2 - 3\lambda + 2) + (\lambda^2 - 4) = 0 \quad \text{is satisfied}$$

by more than two values of x then $\lambda =$

- 1) 1
- 2) -2
- 3) 3
- 4) 2

23. The number of real roots of the equation

$$(\sqrt{3} + 1)^{2x} + (\sqrt{3} - 1)^{2x} = 2^{3x} \quad \text{is equal to}$$

- 1) 0
- 2) 1
- 3) 2
- 4) more than 2

24. If $a \in \mathbb{Z}$ and the equation

$$(x - a)(x - 10) + 1 = 0 \quad \text{has integral roots, then values of 'a' are}$$

- 1) 10, 8
- 2) 12, 10
- 3) 12, 8
- 4) 10, 12

25. The number of real values of x for which

$$\left(\frac{a}{b}\right)^x = x - 2x^2 - 4, \quad \text{where } a, b > 0$$

- 1) 1
- 2) 2
- 3) 0
- 4) infinite

26. Number of solutions of the equation

$$|x| = \cos x$$

- 1) 1
- 2) 2
- 3) 0
- 4) 3

COMMON ROOTS

27. The value of λ in order that the equations $2x^2 + 5\lambda x + 2 = 0$ and $4x^2 + 8\lambda x + 3 = 0$ have a common root is given by

- 1) 1
- 2) -1
- 3) ± 1
- 4) 3

28. If every pair from among the equations $x^2 + ax + bc = 0$, $x^2 + bx + ca = 0$ and $x^2 + cx + ab = 0$ has common root, then the sum of three common roots is

- 1) abc
- 2) $2abc$
- 3) $3(a + b + c)$
- 4) $(a + b + c)$

29. If $x^2 - hx - 21 = 0$, $x^2 - 3hx + 35 = 0$ ($h > 0$) have a common root, then the value of h is

- 1) ± 8
- 2) ± 4
- 3) 4
- 4) 2

30. The conditions that a root of the equation $ax^2 + bx + c = 0$ may be reciprocal to a root of $a_1x^2 + b_1x + c_1 = 0$ is

- 1) $(bb_1 - aa_1)^2 = (ab_1 - bc_1)(ba_1 - b_1c)$
- 2) $(cb_1 - ba_1)^2 = (ac_1 - cd_1)(ab_1 - bc_1)$
- 3) $(cc_1 - aa_1)^2 = (ab_1 - bc_1)(ba_1 - b_1c)$
- 4) $a + b + c = 0$

LOCATION OF ROOTS

31. If both roots of the equation $x^2 + x + a = 0$ exceed 'a' then

- 1) $2 < a < 3$
- 2) $a > 3$
- 3) $-3 < a < 3$
- 4) $a < -2$

32. If both roots of the equation

$$x^2 - 2ax + a^2 - 1 = 0 \quad \text{lie in the interval } (-3, 4)$$

then sum of the integral parts of 'a' is

- 1) 0
- 2) 2
- 3) 4
- 4) -1

33. The expression

$$(a - 2)x^2 + 2(2a - 3)x + (5a - 6) \quad \text{is positive}$$

for all real values of x , then

- 1) 'a' can be any real number
- 2) $a > 1$
- 3) $a > 3$
- 4) $a = 3$

SIGN OF THE EXPRESSIONS AND INEQUALITIES

34. If $x^2 + 2ax + b \geq c \forall x \in \mathbb{R}$ then

- 1) $b - c \geq a^2$
- 2) $c - a \geq b^2$
- 3) $a - b \geq c^2$
- 4) $a + b + c \geq 0$

35. The roots of the equation

$$(x - a)(x - b) + (x - b)(x - c) \\ + (x - c)(x - a) = 0$$

are real and equal if

- 1) $a > b > c$
- 2) $a = b = c$
- 3) $a < b < c$
- 4) $a + b + c = 0$

36. The largest negative integer for which

$$\frac{(x - 4)(x - 2)}{(x - 1)(x - 5)} > 0 \quad \text{is}$$

- 1) -2
- 2) -1
- 3) -3
- 4) -4

**MAXIMUM AND MINIMUM OF
EXPRESSION**

37. Let $f(x) = x^2 + 2bx + 2c^2$ and $g(x) = -x^2 - 2cx + b^2$, then $\min\{f(x)\} > \max\{g(x)\}$ holds for
- 1) $0 < c < b\sqrt{2}$
 - 2) $|c| > |b|\sqrt{2}$
 - 3) $|c| < |b|\sqrt{2}$
 - 4) no real value of b and c

MODULUS FUNCTIONS

38. Number of rational roots of the equation $|x^2 - 2x - 3| + 4x = 0$ is
- 1) 1
 - 2) 2
 - 3) 3
 - 4) 4
39. The roots of the equation $|x^2 - x - 6| = x + 2$ are
- 1) 1, -2, 2
 - 2) 1, 2, 4
 - 3) -2, 2, 4
 - 4) -2, 2, 3
40. For $a > 0$, all the real roots of the equation $x^2 - 3a|x - a| - 7a^2 = 0$ are
- 1) $4a, 5a$
 - 2) $-4a, 5a$
 - 3) $-4a, -5a$
 - 4) $4a, -5a$
41. The set of real values of x satisfying $|x - 1| \leq 3$ and $|x - 1| \geq 1$
- 1) $[2, 4]$
 - 2) $(-\infty, 2) \cup (4, \infty)$
 - 3) $[-2, 0] \cup [2, 4]$
 - 4) $[0, 2]$

OTHER MODELS

42. If $\sin \theta, \sin k\theta$ are the roots of $4x^2 + 2x - 1 = 0$, then k =
- 1) -2
 - 2) -3
 - 3) 3
 - 4) 2
43. If α, β are the roots of $x^2 - 3x + a = 0$ and γ, δ that of $x^2 - 12x + b = 0$ and $\alpha, \beta, \gamma, \delta$ form an increasing G.P. then
- 1) $a = 3, b = 12$
 - 2) $a = 4, b = 16$
 - 3) $a = 2, b = 32$
 - 4) $a = 12, b = 3$
44. Let $f(x)$ be a polynomial for which the remainders when divided by $x - 1, x - 2, x - 3$ respectively 3, 7, 13. Then the remainder of $f(x)$ when divided by $(x - 1)(x - 2)(x - 3)$ is
- 1) $f(x)$
 - 2) $x^2 + x + 1$
 - 3) $x^2 + 1$
 - 4) $x + 2$

45. Let a, b, c be the real numbers, $a \neq 0$, If α is a root of $a^2x^2 + bx + c = 0$, β is a root of $a^2x^2 - bx - c = 0$ and $0 < \alpha < \beta$, then the equation $a^2x^2 + 2bx + 2c = 0$, has a root γ that always satisfies
- 1) $\gamma = (\alpha + \beta) / 2$
 - 2) $\gamma = (\alpha + \beta) / 2$
 - 3) $\gamma = \alpha$
 - 4) $\alpha < \gamma < \beta$
46. The equation $(x - a)^3 + (x - b)^3 + (x - c)^3 = 0$ has
- 1) all the roots are equal
 - 2) one real and two imaginary
 - 3) 3 real roots namely $x = a, x = b, x = c$
 - 4) no real roots

TRANSFORMED EQUATIONS

47. The value of 'a' for which the quadratic equation $3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$ possesses roots of opposite signs lies on
- 1) $(-\infty, 1)$
 - 2) $(-\infty, 0)$
 - 3) $(1, 2)$
 - 4) $[1, 2]$
48. If the roots of the equation $ax^2 + bx + c = 0$ are the reciprocals of the roots of the equation $px^2 + qx + r = 0$ then
- 1) $acq^2 = b^2 pr$
 - 2) $ac = pr$
 - 3) $b^2 ac = q^2 pr$
 - 4) $ab = pq$

RANGE

49. If $y = \tan x \cot 3x, x \in R$ then
- 1) $\frac{1}{3} < y < 1$
 - 2) $\frac{1}{3} \leq y \leq 1$
 - 3) $\frac{1}{3} \leq y \leq 3$
 - 4) $y \leq \frac{1}{3}$ or $y \geq 3$
50. If $5^x + (2\sqrt{3})^{2x} \geq 13^x$ then the solution set for x is
- 1) $[2, +\infty)$
 - 2) $\{2\}$
 - 3) $(-\infty, 2]$
 - 4) $[0, 2]$

INEQUALITIES

51. If $x^2 + 6x - 27 > 0; -x^2 + 3x + 4 > 0$ then x lies in the interval
- 1) $(3, 4)$
 - 2) $\{3, 4\}$
 - 3) $(-\infty, 3) \cup (4, \infty)$
 - 4) $(-9, 4)$

52. The range of values of x which satisfy

$$5x+2 < 3x+8 \text{ and } \frac{x+2}{x-1} < 4 \text{ are}$$

- 1) $(2,3)$ 2) $(-\infty,1) \cup (2,3)$
3) $(2,\infty)$ 4) R

53. The set of exhaustive values of 'a' for which the inequality $(a^2+3)x^2+(a+2)x-4 < 2$ holds of at least one negative x , is

- 1) $a > 0$ 2) $a < 0$
3) $-\infty < a < \infty$ 4) $a \in \phi$

QUADRATIC EXPRESSION IN TWO VARIABLES

54. If the expression

$3x^2+2pxy+2y^2+2ax-4y+1$ can be resolved into two linear factors then p must be a root of the equation

- 1) $x^2+ax+6=0$ 2) $x^2+4ax+6=0$
3) $x^2+4ax+2a^2+6=0$ 4) $x^2-4ax+6=0$

55. The set of values of x for which the inequalities $x^2-3x-10 < 0$, $10x-x^2-16 > 0$ hold simultaneously is (EAM - 2007)

- 1) $(-2,5)$ 2) $(2,8)$ 3) $(-2,8)$ 4) $(2,5)$

56. If p and q are distinct prime numbers and if the equation $x^2-px+q=0$ has positive integers as its roots, then the roots of the equation are (Eamcet-2014)

- 1) 1, -1 2) 2, 3 3) 1, 2 4) 3, 1

57. Let $f(x)=x^2+ax+b$, where $a, b \in R$. If $f(x)=0$ has all its roots imaginary then the roots of $f(x)+f'(x)+f''(x)=0$ are (EAM - 2009)

- 1) real and distinct 2) imaginary
3) equal 4) rational and equal

58. For $x \in R$, the least value of $\frac{x^2-6x+5}{x^2+2x+1}$ is (EAM - 2010)

- 1) -1 2) $-\frac{1}{2}$ 3) $-\frac{1}{4}$ 4) $-\frac{1}{3}$

59. The product of real roots of the equation $|x|^{6/5}-26|x|^{3/5}-27=0$ is (EAM - 2012)

- 1) -3^{10} 2) -3^{12} 3) $-3^{12/5}$ 4) $-3^{21/5}$

60. The set of solutions satisfying both $x^2+5x+6 \geq 0$ and $x^2+3x-4 < 0$ is (EAM - 2013)

- 1) $(-4,1)$ 2) $(-4,-3] \cup [-2,1)$
3) $(-4,-3) \cup (-2,1)$ 4) $[-4,-3] \cup [-2,1]$

61. If α and β are the roots of the equation $x^2-2x+4=0$, then $\alpha^9+\beta^9$ is equal to (EAM - 2013)

- 1) -2^8 2) 2^9 3) -2^{10} 4) 2^{10}

LEVEL-II (H.W.) - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 2 | 2) 3 | 3) 1 | 4) 2 | 5) 2 | 6) 3 |
| 7) 3 | 8) 1 | 9) 2 | 10) 4 | 11) 3 | 12) 4 |
| 13) 2 | 14) 1 | 15) 4 | 16) 3 | 17) 1 | 18) 1 |
| 19) 4 | 20) 1 | 21) 3 | 22) 4 | 23) 2 | 24) 3 |
| 25) 3 | 26) 2 | 27) 3 | 28) 4 | 29) 3 | 30) 3 |
| 31) 4 | 32) 1 | 33) 3 | 34) 1 | 35) 2 | 36) 2 |
| 37) 2 | 38) 1 | 39) 3 | 40) 4 | 41) 3 | 42) 2 |
| 43) 3 | 44) 2 | 45) 4 | 46) 2 | 47) 3 | 48) 1 |
| 49) 4 | 50) 3 | 51) 1 | 52) 1 | 53) 3 | 54) 3 |
| 55) 4 | 56) 3 | 57) 2 | 58) 4 | 59) 1 | 60) 2 |
| 61) 3 | | | | | |

LEVEL-II (H.W.) - HINTS

1. $\alpha + \beta = \frac{3}{8}, \alpha\beta = \frac{27}{8}$

$$\left(\frac{\alpha^2}{\beta}\right)^{1/3} + \left(\frac{\beta^2}{\alpha}\right)^{1/3} = \frac{\alpha + \beta}{(\alpha\beta)^{1/3}} = \frac{3/8}{3/2} = \frac{1}{4}$$

2. $m = \alpha\beta + \alpha + \beta = q + p,$

$$n = \alpha\beta - \alpha - \beta = q - p$$

The quadratic equation which roots m and n

$$\text{is } x^2 - 2qx + q^2 - p^2 = 0$$

3. $\sin \theta + \cos \theta = -b/a; \sin \theta \cos \theta = c/a$

$$1 + 2\frac{c}{a} = \frac{b^2}{a^2}; a^2 + 2ac = b^2$$

4. $\alpha = -\omega, \beta = -\omega^2; \alpha^5 + \beta^5 = -\omega^2 - \omega = 1$

5. $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2} = \frac{b^2-2ac}{a^2c^2}$
 $= \frac{25+42}{9 \times 49} = \frac{67}{441}$
6. $\alpha + \beta = \alpha^2 + \beta^2$; $\frac{-b}{a} = \frac{b^2-2ac}{a^2}$
 $\therefore b^2 + ab = 2ac$
7. given expression =
 $= a + \frac{b}{2}(\alpha + \beta) + \frac{c}{2}(\alpha^2 + \beta^2) + \frac{d}{2}(\alpha^3 + \beta^3)$
 $= a + \frac{b}{2}(\alpha + \beta) + \frac{c}{2}[(\alpha + \beta)^2$
 $- 2\alpha\beta] + \frac{d}{2}(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$
 $= a + \frac{b}{2}(1) + \frac{c}{2}[(1)^2 - 2(1/6)]$
 $+ \frac{d}{2}[(1)^3 - 3(1/6)(1)] = a + \frac{b}{2} + \frac{c}{3} + \frac{d}{4}$
8. $\alpha + \beta = -a$, $\alpha\beta = -b$; $\gamma + \delta = -a$, $\gamma\delta = b$
 $(\alpha^2 + a\alpha + b)(\beta^2 + a\beta + b) = 4b^2$
9. $\frac{\alpha + \beta}{2} = 8/5$; $\alpha + \beta = 16/5$
 $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{16}{7}$; $\frac{16}{5} \times \frac{7}{16} = \alpha\beta$ $\alpha\beta = 7/5$
 The equation is $5x^2 - 16x + 7 = 0$
10. $a^2c + ac^2 = 3abc - b^3$
11. $\frac{(m+n)^2}{mn} = \frac{4b^2}{ac} = \frac{4q^2}{pr} \Rightarrow \frac{b^2}{ac} = \frac{q^2}{pr}$
12. $|\alpha - \beta|^2 = |\gamma - \delta|^2$
 $\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = (\gamma + \delta)^2 - 4\gamma\delta$
 $\Rightarrow b^2 - 4c = c^2 - 4b$
 $b^2 - c^2 = 4c - 4b \Rightarrow b + c = -4$
13. $\frac{\alpha + \beta}{2} = a$; $\alpha_1\beta_1 = a^2 \Rightarrow \sqrt{\alpha_1\beta_1} = a$
 $\therefore$ A.M of roots of 2nd equation = G.M of roots of 1st equation
14. $\tan A + \tan B = 2$; $\tan A \tan B = 2$

- $\tan(A+B) = \frac{2}{1-2} = -2$, $\sin^2(A+B) = \frac{4}{5}$
15. Coefficient of $x=0$
16. $\Delta_1 + \Delta_2 = p^2 - 4q + r^2 - 4s$
 $= p^2 + r^2 - 4(q+s) = p^2 + r^2 - 2pr = (p-r)^2 \geq 0$
 at least one of $\Delta_1 \geq 0, \Delta_2 \geq 0$
 $\therefore$ at least one has real roots.
17. $\Delta = 4(c-a)^2 - 4(b-c)(a-b)$
 $= 4[c^2 + a^2 - 2ac - ab + b^2 + ac - bc]$
 $= 4[a^2 + b^2 + c^2 - ab - bc - ca] > 0$
18. $\Delta = 4c^2 - 4(b+c-a)(a+c-b)$
 $= 4[c^2 - [c - (a-b)][c + a - b]]$
 $= 4(a-b)^2 \geq 0$ and $a \neq b$ roots are real and distinct
19. $c^2 - ab > 0$ then
 $\Delta = 4[(a+b)^2 - (a^2 + b^2 + 2c^2)]$
 $= 4[2ab - 2c^2] < 0$
20. If Δ_1, Δ_2 and Δ_3 be the discriminants of individual quadratics, then $D_1 + D_2 + D_3 > 0$
21. $\Delta = 4(a+c)^2 - 4(a+c+b)(a+c-b)$
 $\Delta = 4(a+c)^2 - 4[(a+c)^2 - b^2]$; $\Delta = 4b^2$
22. $a=b=c=0$
23. The equation is $\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)^{2x} + \left(\frac{\sqrt{3}-1}{2\sqrt{2}}\right)^{2x} = 1$
 $\Rightarrow (\sin \theta)^{2x} + (\cos \theta)^{2x} = 1$
24. Since 'a' and 'x' are integers
 $\therefore (x-a)(x-10)+1=0 \Rightarrow (x-a)(x-10)=-1$
 $\Rightarrow (x-a=1, x-10=-1)$
 or $(x-a=-1 \text{ and } x-10=1)$
 $\Rightarrow (x=9, a=8) \text{ or } (x=11 \text{ and } a=12)$
 Thus $a=8 \text{ or } 12$

25. $LHS > 0$; $RHS = -2\left(x^2 - \frac{x}{2}\right) - 4$
- $$= -2\left[\left(x - \frac{1}{4}\right)^2 - \frac{1}{16}\right] - 4$$
- $$= -2\left[\left(x - \frac{1}{4}\right)^2\right] - \frac{31}{8} < 0$$
- given equation has no solution.
Number of solutions is 0
26. draw the graph
27. $(a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) = (c_1a_2 - c_2a_1)^2$
 $\therefore 4\lambda^2 = 4 \Rightarrow \lambda = \pm 1$
28. Common root of $x^2 + ax + bc = 0$ and $x^2 + bx + ca = 0$ is 'c' common root of $x^2 + bx + ca = 0$ and $x^2 + cx + ab = 0$ is 'a' common root of $x^2 + cx + ab = 0$ and $x^2 + ax + bc = 0$ is 'b'.
29. $\alpha^2 - h\alpha - 21 = 0$ and $\alpha^2 - 3h\alpha + 35 = 0$
 $\Rightarrow 2h\alpha - 56 = 0 \Rightarrow \alpha = \frac{28}{h}$
 Substitute in either of the equations
30. $ax^2 + bx + c = 0$ and $c_1x^2 + b_1x + a_1 = 0$ have a common root then find the condition
31. $f(a) > 0$ and $\alpha + \beta > 2a$ and $\Delta > 0$
32. $x^2 - 2ax + a^2 - 1 = 0 \Rightarrow (x - a)^2 = 1$
 Roots are $a - 1$ and $a + 1$
 Both roots lie in $(-3, 4) \therefore a - 1 > -3, a + 1 < 4$
 ie $-2 < a < 3 \Rightarrow [a] = -2, -1, 0, 1, 2$ whose sum is zero.
33. $\Delta < 0 \Rightarrow (2a - 3)^2 - (a - 2)(5a - 6) < 0$
 $4a^2 - 12a + 9 - 5a^2 + 16a - 12 < 0$
 $\Rightarrow -a^2 + 4a - 3 < 0 \Rightarrow a^2 - 4a + 3 > 0$
 $\Rightarrow a \in (-\infty, 1) \cup (3, \infty)$ and $a - 2 > 0$
34. $x^2 + 2ax - b - c \geq 0 \forall x \in R \Rightarrow \Delta \leq 0$
35. $\Delta = 4\left((a + b + c)^2 - (ab + bc + ca)\right)$
 $= 2\left((a - b)^2 + (b - c)^2 + (c - a)^2\right) = 0$

36. Let $y = \frac{(x-4)(x-2)}{(x-1)(x-5)}$
 $\therefore x < 1$ or $2 < x < 4$ or $x > 5$
37. $\min\{f(x)\} = 2c^2 - b^2$ and
 $\max\{g(x)\} = b^2 + c^2$
38. use extreme value formula
39. by verification
40. by verification
41. $|x - 1| \leq 3 \Rightarrow -3 \leq x - 1 \leq 3$
 $\Rightarrow -2 \leq x \leq 4$ and $|x - 1| \geq 1$
 $\Rightarrow x - 1 \leq -1$ or $x - 1 \geq 1$
 $\Rightarrow x \leq 0$ or $x \geq 2$ combining $x \in [-2, 0] \cup [2, 4]$
42. $\alpha, \beta = \frac{-2 \pm \sqrt{4 + 16}}{8} = \frac{\pm\sqrt{5} - 1}{4}$
 $\alpha = \sin 18^\circ, \beta = -\left(\frac{\sqrt{5} + 1}{4}\right) = \sin 54^\circ = \sin(-54^\circ) \Rightarrow$
 $\theta = 18^\circ; k = -3$
43. by verification
44. Verification
45. α is a root of $a^2x^2 + bx + c = 0$
 $\Rightarrow a^2\alpha^2 + b\alpha + c = 0$
 β is a root of $a^2x^2 - bx - c = 0$
 $\Rightarrow a^2\beta^2 - b\beta - c = 0$
 Let $f(x) = a^2x^2 + 2bx + 2c$
 $f(\alpha) = a^2\alpha^2 + 2b\alpha + 2c = -a^2\alpha^2 < 0$
 $f(\beta) = a^2\beta^2 + 2b\beta + 2c = 3a^2\beta^2 > 0$
 by rolles theorem $\exists \gamma \in (\alpha, \beta) \ni f(\gamma) = 0$
46. $f(x) = (x - a)^3 - (x - b)^3 + (x - c)^3$
 $f'(x) = 3(x - a)^2 + 3(x - b)^2 + 3(x - c)^2 > 0$
 $\forall x \in R$ $f(x)$ is strictly increasing
 $\therefore f(x) = 0$ has one real and two imaginary
47. $\alpha\beta < 0 \Rightarrow a^2 - 3a + 2 < 0; a \in (1, 2)$

48. $cx^2 + bx + a = 0$; $px^2 + qx + r = 0$ have same roots

$$\therefore \frac{c}{p} = \frac{b}{q} = \frac{a}{r}$$

$$\therefore ac / pr = b^2 / q^2$$

49. $y = \frac{\tan x(1 - 3 \tan^2 x)}{3 \tan x - \tan^3 x} = \frac{1 - 3t^2}{3 - t^2}$

$$(y - 3)t^2 - 3y + 1 = 0 \Rightarrow \Delta \geq 0$$

50. $\therefore \cos^x \alpha + \sin^x \alpha \geq 1$,

where $\cos \alpha = 5/13$ equality holds for $x=2$,

if $x < 2$ both $\cos \alpha$ and $\sin \alpha$ increase
 $\cos^x \alpha + \sin^x \alpha > 1$ if $x < 2$

51. $x^2 + 6x - 27 > 0 \Rightarrow (x + 9)(x - 3) > 0$

$$\Rightarrow x < -9 \text{ or } x > 3$$

also, $-x^2 + 3x + 4 > 0 \Rightarrow (x - 4)(x + 1) < 0$

$$\Rightarrow -1 < x < 4 \dots (2)$$

From (1) and (2) $x \in (3, 4)$

52. $5x + 2 < 3x + 8 \Rightarrow x < 3 \dots (1)$

$$\frac{x+2}{x-1} < 4 \Rightarrow \frac{x+2}{x-1} - 4 < 0$$

$$\Rightarrow \frac{x-2}{x-1} > 0 \Rightarrow x < 1 \text{ or } x > 2 \dots (2)$$

From (1) and (2) $x \in (-\infty, 1) \cup (2, 3)$

53. If $f(x) = (a^2 + 3)x^2 + (a + 2)x - 6$,

then $f(0) < 0$

54. $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

55. Solve the inequalities find the common interval.

56. let $p = 3$ and $q = 2$, then given equation becomes
 $x^2 - 3x + 2 = 0$

57. $f(x) = x^2 + ax + b$ has imaginary roots

$$\Delta < 0 \Rightarrow a^2 - 4b < 0$$

$$f'(x) = 2x + a;$$

$$f''(x) = 2$$

$$\Rightarrow x^2 + (a + 2)x + b + a + 2 = 0$$

$$\therefore x = \frac{-(a + 2) \pm \sqrt{(a + 2)^2 - 4(a + b + 2)}}{2}$$

$$\therefore a^2 - 4b < 0$$

$$a^2 - 4a - 4 < 0$$

$\therefore$ equation has imaginary roots

58. Let $y = \frac{x^2 - 6x + 5}{x^2 + 2x + 1}$

$$(y - 1)x^2 + (2y + 6)x + (y - 5) = 0$$

$$x \in R, \Delta \geq 0$$

59. Put $|x|^{3/5} = t$

$$\therefore t^2 - 26t - 27 = 0 \Rightarrow t = 27 \text{ or } -1$$

$$|x|^{3/5} = 27$$

$$\therefore |x|^{3/5} \text{ cannot be negative so } x = 3^5 \text{ or } -3^5$$

$$\text{Product } x = 3^5 \times (-3)^5 = -3^{10}$$

60. Verification

61. $\alpha + \beta = 2;$

$$\alpha\beta = 4$$

$$\alpha^9 + \beta^9 = (\alpha^3)^3 + (\beta^3)^3$$



LEVEL-III

1. If $0 < \alpha < \beta < \gamma < \frac{\pi}{2}$, then the equation

$$\frac{1}{x - \sin \alpha} + \frac{1}{x - \sin \beta} + \frac{1}{x - \sin \gamma} = 0 \text{ has}$$

- 1) imaginary roots 2) real and equal roots
3) real and unequal roots 4) rational roots

2. If n is a multiple of 6 and α, β are the roots

of $x^2 + x + 1 = 0$ then $(1 + \alpha)^{-n} + (1 + \beta)^{-n} =$

- 1) 2 2) -2 3) 0 4) -1

3. If α, β are real and $\alpha^2, -\beta^2$ are the roots of

the equation $a^2x^2 + x + (1 - a^2) = 0 (a > 1)$ then $\beta^2 =$

- 1) a^2 2) 1 3) $1 - a^2$ 4) $1 + a^2$

4. If α, β are the roots of $x^2 + px - q = 0$ and

γ, δ are that of $x^2 + px + r = 0$ then

$$(\alpha - \gamma)(\beta - \gamma)(\alpha - \delta)(\beta - \delta) =$$

1) $(q - r)^2$ 2) $(q + r)^2$

3) $-(q + r)^2$ 4) $-(q - r)^2$

5. If α, β are the roots of $ax^2 + bx + c = 0$ and

$S_n = \alpha^n + \beta^n$ then $aS_{n+1} + bS_n + cS_{n-1} =$

- 1) 0 2) 1 3) 2 4) 5

6. α and β are the roots of the equation

$x^2 + px + p^3 = 0, (p \neq 0)$. If the point (α, β)

lies on the curve $x = y^2$ then the roots of the given equation are

- 1) 4, -2 2) 4, 2 3) 1, -1 4) 1, 1

7. If a and b are distinct positive real numbers

such that $a, a_1, a_2, a_3, a_4, a_5, b$ are in A.P;

$a, b_1, b_2, b_3, b_4, b_5, b$ are in GP; and

$a, c_1, c_2, c_3, c_4, c_5, b$ are in HP; Then the roots

of $a_3x^2 + b_3x + c_3 = 0$ are

- 1) real and distinct 2) real and equal

- 3) imaginary 4) rational

8. p, q, r and s are integers. If the A.M. of the

roots of $x^2 - px + q^2 = 0$ and G.M. of the roots

of $x^2 - rx + s^2 = 0$ are equal then

- 1) q is an odd integer 2) r is an even integer

- 3) p is an even integer 4) s is an odd integer

9. If $x^2 + x + 1 = 0$ then the value of

$$\left(x + \frac{1}{x}\right) + \left(x^2 + \frac{1}{x^2}\right) + \dots + \left(x^{52} + \frac{1}{x^{52}}\right) \text{ equals}$$

- 1) 1 2) 0 3) -1 4) -52

10. The value of the continued fraction

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots \infty}}} \text{ is}$$

- 1) $\frac{\sqrt{5}-1}{2}$ 2) $\frac{\sqrt{5}+1}{2}$ 3) $\frac{\sqrt{5}-1}{4}$ 4) $\frac{\sqrt{5}+1}{4}$

11. Number of real roots of the equation

$$\sqrt{x} + \sqrt{x - \sqrt{1-x}} = 1 \text{ is}$$

- 1) 0 2) 1 3) 2 4) 3

12. The equation

$$2 \cos^2 \frac{x}{2} \sin^2 x = x^2 + \frac{1}{x^2}, 0 \leq x \leq \frac{\pi}{2} \text{ has}$$

- 1) One real solution

- 2) No real solution

- 3) More than one real solution

- 4) two real solutions

13. The quadratic equations $x^2 - 6x + a = 0$ and

$x^2 - cx + 6 = 0$ have one root common. The other roots of the first and second equations are integers in the ratio 4:3. Then the common root is (AIEEE 2008)

- 1) 3 2) 2 3) 1 4) 4

14. If the two equations $x^2 - cx + d = 0$ and $x^2 - ax + b = 0$ have a common root and the second equation has equal roots then

1) $b + d = ac$ 2) $2(b + d) = ac$

3) $b + d = 2ac$ 4) $(b + d)^2 = a + c$

15. If $x^2 + 2ax + a < 0 \quad \forall x \in [1, 2]$ then

1) $a \in \left(-\infty, -\frac{4}{5}\right)$ 2) $a \in (0, \infty)$

3) $a \in \left(-\frac{4}{5}, 0\right)$ 4) $a \in \left(\frac{4}{5}, \infty\right)$

16. If $f(x)$ is a quadratic expression which is positive for all real values of x and

$g(x) = f(x) + f'(x) + f''(x)$ then for any real value of x

1) $g(x) < 0$ 2) $g(x) > 0$ 3) $g(x) = 0$ 4) $g(x) = 2$

17. If the roots of the equation $x^2 + 2ax + b = 0$ are real and distinct and they differ by at most $2m$ then 'b' lies in the interval

1) $(a^2 - m^2, a^2)$ 2) $[a^2 - m^2, a^2)$
3) $(a^2, a^2 + m^2)$ 4) $[a^2 + m^2, a^2)$

18. The solution of the equation $(3|x| - 3)^2 = |x| + 7$ which belongs to the domain of $\sqrt{x(x-3)}$ are given by

1) $\pm \frac{1}{9}, \pm 2$ 2) $\frac{1}{9}, -2$ 3) $-\frac{1}{9}, -2$ 4) $-\frac{1}{9}, 2$

19. For $a \leq 0$, the roots of the equation $x^2 - 2a|x - a| - 3a^2 = 0$ are IIT-1987

1) $(1 - \sqrt{2})a, (-1 + \sqrt{6})a$
2) $(1 + \sqrt{2})a, (-1 + \sqrt{6})a$
3) $(1 + \sqrt{2})a, (-1 - \sqrt{6})a$
4) $(1 - \sqrt{2})a, (-1 - \sqrt{6})a$

20. If n is a positive integer and $n \in [5, 100]$ then the number of positive integral roots of the equation $x^2 + 2x - n = 0$ is

1) 4 2) 6 3) 8 4) 10

21. The number of solutions of the system of equations given below is

$$|x| + |y| = 1; x^2 + y^2 = a^2; \left(\frac{1}{\sqrt{2}} < a < 1 \right)$$

1) infinite 2) 2 3) 4 4) 8

22. If α, β are the roots of $x^2 + x + 1 = 0$ and

$$S_n = \alpha^n + \beta^n \text{ then } \begin{vmatrix} 3 & 1 + S_1 & 1 + S_2 \\ 1 + S_1 & 1 + S_2 & 1 + S_3 \\ 1 + S_2 & 1 + S_3 & 1 + S_4 \end{vmatrix} =$$

1) 27 2) -27 3) -3 4) 9

23. If α, β are the roots of the equation $\lambda(x^2 - x) + x + 5 = 0$ and if λ_1 and λ_2 are the two values of λ for which the roots α, β

are connected by the relation $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{4}{5}$, then

the value of $\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1} =$

1) 250 2) 1022 3) 254 4) 0

24. If $0 < x < 1000$ and $\left[\frac{x}{2} \right] + \left[\frac{x}{3} \right] + \left[\frac{x}{5} \right] = \frac{31}{30}x$

where $[x]$ is the greatest Integer less than or equal to x , then the number of possible values of x is

1) 34 2) 32 3) 33 4) 35

25. If α, β are the roots of the equation $ax^2 + bx + c = 0$ then the value of

$(1 + \alpha + \alpha^2)(1 + \beta + \beta^2)$ is

1) positive 2) negative
3) non-negative 4) non positive

26. The equation

$$(x-3)^9 + (x-3^2)^9 + (x-3^3)^9 + \dots + (x-3^9)^9 = 0 \text{ has}$$

1) all the roots are real
2) one real and 8 imaginary roots
3) real roots namely $x = 3, 3^2, \dots, 3^9$
4) five real and 4 imaginary roots

27. The root of the equation

$$2(1+i)x^2 - 4(2-i)x - 5 - 3i = 0, \text{ where}$$

$i = \sqrt{-1}$, which has greater modulus is

1) $(3-5i)/2$ 2) $(5-3i)/2$
3) $(3+i)/2$ 4) $(1+3i)/2$

28. If $xy = 2(x+y), x \leq y$ and $x, y \in N$, then number of solutions of the equation

1) two 2) three
3) no solution 4) infinitely many solution

29. If $c > 0$ and the equation $3ax^2 + 4bx + c = 0$ has no real root, then

1) $2a + c > b$ 2) $a + 2c > b$
3) $3a + c > 4b$ 4) $a + 3c < b$

30. The polynomial

$(ax^2 + bx + c)(ax^2 - dx - c), ac \neq 0$, has

- 1) four real zeroes
- 2) at least two real zeroes
- 3) at most two real zeros
- 4) all the above

31. The range of a for which the equation $x^2 + ax - 4 = 0$ has its smaller root in the interval $(-1, 2)$ is

- 1) $(-\infty, -3)$
- 2) $(0, 3)$
- 3) $(0, \infty)$
- 4) $(-\infty, -3) \cup (0, \infty)$

32. If x, y, z are real and distinct, then $x^2 + 4y^2 + 9z^2 - 6yz - 3zx - 2xy$ is always

- 1) positive
- 2) non negative
- 3) negative
- 4) zero

33. If $\sqrt{9x^2 + 6x + 1} < 2 - x$ then

- 1) $x \in \left(-\frac{3}{2}, \frac{1}{4}\right)$
- 2) $x \in \left(-\frac{3}{2}, \frac{1}{4}\right]$
- 3) $x \in \left[-\frac{3}{2}, \frac{1}{4}\right)$
- 4) $x < \frac{1}{4}$

34. For the quadratic equation

$$4x^2 - 2(a + c - 1)x + ac - b = 0 \quad (a > b > c)$$

- 1) both roots are greater than a
- 2) both roots are less than c
- 3) both roots lie between $c/2$ and $a/2$
- 4) exactly one of the roots lies between $c/2$ and $a/2$

35. If $\frac{2x}{2x^2 + 5x + 2} > \frac{1}{x+1}$, then

- 1) $x < 2$
- 2) $-1 < x < 1$
- 3) $-2 < x < 0$
- 4) $-\frac{2}{3} < x < -\frac{1}{2}$

36. If x, y, z are real such that $x + y + z = 4$, $x^2 + y^2 + z^2 = 6$, then $x \in$

- 1) $(-1, 1)$
- 2) $[0, 2]$
- 3) $[2, 3]$
- 4) $\left[\frac{2}{3}, 2\right]$

37. The equation $2^{2x} + (a-1)2^{x+1} + a = 0$ has roots of opposite signs then exhaustive set of values of a is

- 1) $a \in (-1, 0)$
- 2) $a < 0$
- 3) $a \in (-\infty, 1/3)$
- 4) $a \in (0, 1/3)$

38. If the equation $|x^2 + bx + c| = k$ has four real roots, then

- 1) $b^2 - 4c > 0$ and $0 < k < \frac{b^2 - 4c}{4}$
- 2) $b^2 - 4c < 0$ and $0 < k < \frac{b^2 - 4c}{4}$
- 3) $b^2 - 4c > 0$ and $k > \frac{b^2 - 4c}{4}$
- 4) $b^2 - 4c < 0$ and $k > \frac{b^2 - 4c}{4}$

39. The set of values of a for which

$(a-1)x^2 - (a+1)x + a - 1 \geq 0$ is true for all $x \geq 2$

- 1) $(-\infty, 1)$
- 2) $\left(1, \frac{7}{3}\right]$
- 3) $\left(\frac{7}{3}, \infty\right)$
- 4) $(1, \infty)$

40. If α, β are the roots of $x^2 + px + q = 0$ and

$x^{2n} + p^n x^n + q = 0$ and if $\left(\frac{\alpha}{\beta}\right), \left(\frac{\beta}{\alpha}\right)$ are the

roots of $x^n + 1 + (x+1)^n = 0, (n \in N)$ then

- 1) must be an odd integer
- 2) may be any integer
- 3) must be an even integer
- 4) cannot say anything

41. If $(x^2 + x + 2)^2 - (a-3)(x^2 + x + 1)(x^2 + x + 2)$

$+ (a-4)(x^2 + x + 1)^2 = 0$ has at least one real root, then the complete set of values of a is.

- 1) $\left(5, \frac{19}{3}\right]$
- 2) $\left(\frac{19}{3}, \infty\right)$
- 3) $(-\infty, 5)$
- 4) $(1, 10)$

42. Let α, β be the roots of the equation

$$x^2 - px + r = 0 \text{ and } \frac{\alpha}{2}, 2\beta \text{ be the roots of the}$$

equation $x^2 - qx + r = 0$. Then, the value of r is (AIE - 2007)

- 1) $\frac{2}{9}(p-q)(2q-p)$
- 2) $\frac{2}{9}(q-p)(2p-q)$
- 3) $\frac{2}{9}(q-2p)(2q-p)$
- 4) $\frac{2}{9}(2p-q)(2q-p)$

43. Let p and q be real numbers such that $p \neq 0$, $p^3 \neq q$ and $p^3 \neq -q$. If α and β are non-zero complex numbers satisfying $\alpha + \beta = -p$ and $\alpha^3 + \beta^3 = q$, then a quadratic equation having $\frac{\alpha}{\beta}$ and $\frac{\beta}{\alpha}$ as its roots is (AIE - 2010)

- 1) $(p^3 + q)x^2 - (p^3 + 2q)x + (p^3 + q) = 0$
- 2) $(p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$
- 3) $(p^3 - q)x^2 - (5p^3 - 2q)x + (p^3 - q) = 0$
- 4) $(p^3 - q)x^2 - (5p^3 + 2q)x + (p^3 - q) = 0$

44. Let α and β be the roots of $x^2 - 6x - 2 = 0$, with $\alpha > \beta$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{2a_9}$ is ((AIE - 2011)

- 1) 1
- 2) 2
- 3) 3
- 4) 4

45. A value of b for which the equations $x^2 + bx - 1 = 0$, $x^2 + x + b = 0$ have one root in common is

- 1) $-\sqrt{2}$
- 2) $-i\sqrt{3}$
- 3) $i\sqrt{5}$
- 4) $\sqrt{2}$

46. The real number k for which the equation $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$ (JEE MAIN - 2013)

- 1) lies between 1 and 2
- 2) lies between 2 and 3
- 3) lies between -1 and 0
- 4) does not exist

47. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$, $a, b, c \in R$, have a common root then $a : b : c$ is (JEE MAIN- 2013)

- 1) 1 : 2 : 3
- 2) 3 : 2 : 1
- 3) 1 : 3 : 2
- 4) 3 : 1 : 2

48. If $a \in R$ and the equation

$-3(x - [x])^2 + 2(x - [x]) + a^2 = 0$ (where, $[x]$ denotes the greatest integer $\leq x$) has no integral solution, then all possible values of a lie in the interval (JEE - 2014)

- 1) $(-1, 0) \cup (0, 1)$
- 2) $(1, 2)$
- 3) $(-2, -1)$
- 4) $(-\infty, -2) \cup (2, \infty)$

49. Let α and β be the roots of the equation $px^2 + qx + r = 0$, $p \neq 0$. If p, q and r are in AP and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is ((JEE MAIN - 2014)

- 1) $\frac{\sqrt{61}}{9}$
- 2) $\frac{2\sqrt{17}}{9}$
- 3) $\frac{\sqrt{34}}{9}$
- 4) $\frac{2\sqrt{13}}{9}$

50. $\left\{x \in R : \frac{14x}{x+1} - \frac{9x-30}{x-4} < 0\right\}$ is equal to (EAM - 2010)

- 1) $(-1, 4)$
- 2) $(1, 4) \cup (5, 7)$
- 3) $(1, 7)$
- 4) $(-1, 1) \cup (4, 6)$

LEVEL - III - KEY

- | | | | | | |
|-------|-------|-------|-------|-------|-------|
| 1) 3 | 2) 1 | 3) 2 | 4) 2 | 5) 1 | 6) 1 |
| 7) 3 | 8) 3 | 9) 3 | 10) 2 | 11) 2 | 12) 2 |
| 13) 2 | 14) 2 | 15) 1 | 16) 2 | 17) 2 | 18) 3 |
| 19) 1 | 20) 3 | 21) 4 | 22) 2 | 23) 3 | 24) 3 |
| 25) 3 | 26) 2 | 27) 1 | 28) 1 | 29) 3 | 30) 2 |
| 31) 1 | 32) 2 | 33) 1 | 34) 4 | 35) 4 | 36) 4 |
| 37) 3 | 38) 1 | 39) 2 | 40) 3 | 41) 1 | 42) 4 |
| 43) 2 | 44) 3 | 45) 2 | 46) 4 | 47) 1 | 48) 1 |
| 49) 4 | 50) 4 | | | | |

LEVEL - III - HINTS

1. $0 < \alpha < \beta < \gamma < \frac{\pi}{2} \Rightarrow \sin \alpha < \sin \beta < \sin \gamma$
 given equation is $f(x) = (x - \sin \beta)(x - \sin \gamma) + (x - \sin \alpha)(x - \sin \gamma) + (x - \sin \alpha)(x - \sin \beta)$
 $f(\sin \alpha) = (\sin \alpha - \sin \beta)(\sin \alpha - \sin \gamma) > 0$
 $f(\sin \beta) = (\sin \beta - \sin \alpha)(\sin \beta - \sin \gamma) < 0$
 $f(\sin \gamma) = (\sin \gamma - \sin \alpha)(\sin \gamma - \sin \beta) > 0$
 $\therefore f(x) = 0$ has one root between $\sin \alpha$ and $\sin \beta$ and another root between $\sin \beta$ and $\sin \gamma$
2. $\alpha = \omega, \beta = \omega^2$
3. $\alpha^2 + (-\beta^2) = -\frac{1}{a^2}, \alpha^2(-\beta^2) = \frac{1-a^2}{a^2} = \frac{1}{a^2} - 1$

$$\alpha^2 \beta^2 = 1 - \frac{1}{a^2} ; \alpha^2 \beta^2 = 1 + \alpha^2 - \beta^2$$

$$\therefore (1 + \alpha^2)(1 - \beta^2) = 0$$

$$4. (\gamma^2 - (\alpha + \beta)\gamma + \alpha\beta)(\delta^2 - (\alpha + \beta)\delta + \alpha\beta) \\ = (\gamma^2 + p\gamma - q)(\delta^2 + p\delta - q)$$

$$5. ax^2 + bx + c = 0$$

$$\Rightarrow ax^{n+1} + bx^n + cx^{n-1} = 0$$

$$a\alpha^{n+1} + b\alpha^n + c\alpha^{n-1} = 0$$

$$a\beta^{n+1} + b\beta^n + c\beta^{n-1} = 0$$

adding above equations

$$\Rightarrow aS_{n+1} + bS_n + cS_{n-1} = 0$$

$$6. \alpha + \beta = -p, \alpha\beta = p^3$$

$$(\alpha, \beta) \text{ lies on } x = y^2 \Rightarrow \alpha = \beta^2$$

$$\Rightarrow \alpha\beta = \beta^3 \Rightarrow \beta^3 = p^3$$

$$\Rightarrow \beta = p \text{ and } \alpha = -2p$$

$$7. (GM)^2 = (AM) \cdot (HM) \Rightarrow b_3^2 = a_3 \cdot c_3$$

when a_3, b_3, c_3 are AM, GM, HM of a and b

$$\text{respectively } \Delta = b_3^2 - 4a_3c_3 = b_3^2 - 4b_3^2 < 0$$

$$8. \text{ By given condition } \frac{\alpha + \beta}{2} = \sqrt{\gamma\delta} \Rightarrow p = 2s.$$

$\therefore p$ is even

$$9. 1 + x + x^2 = 0 \Rightarrow x = w, \frac{1}{w}$$

$$10. x = 1 + \frac{1}{x}$$

$$11. x - \sqrt{1-x} = 1 + x - 2\sqrt{x} ; 1 - x = 1 + 4x - 4\sqrt{x}$$

$$-5x = -4\sqrt{x} \Rightarrow 25x^2 = 16x \Rightarrow x = \frac{16}{25}$$

$$12. L.H.S < 2 \text{ and } R.H.S. \geq 2 ;$$

$$13. \alpha, 4\beta \text{ are roots of } x^2 - 6x + a = 0 \text{ and}$$

$$\alpha, 3\beta \text{ are roots of } x^2 - cx + 6 = 0$$

$$\alpha + 4\beta = 6 \text{ and } 4\alpha\beta = a$$

$$\alpha + 3\beta = c \text{ and } 3\alpha\beta = 6$$

$$\Rightarrow a = 8 \Rightarrow \alpha = 2$$

$$14. x^2 - ax + b = 0 ; \alpha = \frac{a}{2}, \alpha^2 = b ; \text{ put } \alpha \text{ in}$$

$$x^2 - cx + d = 0 ; \alpha^2 - c\alpha + d = 0 ; b + d = \frac{ac}{2}$$

$$15. f(1) < 0 \text{ and } f(2) < 0$$

$$16. f(x) = x^2 + 1, g(x) = x^2 + 2x + 3$$

$$g(x) = (x+1)^2 + 2 > 0$$

$$17. \Delta > 0 \Rightarrow a^2 > b \Rightarrow b < a^2 ; |\alpha - \beta| \leq 2m$$

$$a^2 - b \leq m^2 \Rightarrow a^2 - m^2 \leq b ; b \geq a^2 - m^2$$

18. By verification.

$$19. (i) \text{ suppose } x - a > 0$$

$$x^2 - 2a|x-a| - 3a^2 = 0 \Rightarrow x^2 - 2a(x-a) - 3a^2 = 0$$

$$\Rightarrow x = (1 + \sqrt{2})a$$

$$(ii) \text{ suppose } x - a < 0$$

$$x^2 - 2a|x-a| - 3a^2 = 0 \Rightarrow x^2 + 2a(x-a) - 3a^2 = 0$$

$$\Rightarrow x = (-1 + \sqrt{6})a$$

$$20. x = -1 \pm \sqrt{n+1} \Rightarrow n+1 \text{ must be a perfect square}$$

21. every line intersect the circle in two distinct points.

$$22. \alpha = w, \beta = w^2$$

$$23. \lambda x^2 + (1 - \lambda)x + 5 = 0$$

$$\frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{4}{5} \Rightarrow \frac{(1 - \lambda)^2 - 10\lambda}{\lambda^2 5 / \lambda} = \frac{4}{5}$$

$$\lambda^2 - 12\lambda + 1 = 4\lambda ; \lambda^2 - 16\lambda + 1 = 0$$

$$\frac{\lambda_1}{\lambda_2} + \frac{\lambda_2}{\lambda_1} = \frac{b^2 - 2ac}{ac} = 256 - 2 = 254$$

$$24. \left[\frac{x}{2}\right] + \left[\frac{x}{3}\right] + \left[\frac{x}{5}\right] = \frac{31}{30}x = \frac{x}{2} + \frac{x}{3} + \frac{x}{5}$$

$$\Rightarrow \frac{x}{2}, \frac{x}{3}, \frac{x}{5} \text{ are all integers.}$$

$$\Rightarrow x \text{ is a multiple of '30'}$$

$$25. 1 + \alpha + \alpha^2 \geq 0 \text{ because } \Delta < 0$$

$$1 + \beta + \beta^2 \geq 0 \text{ because } \Delta_1 < 0$$

$$\therefore (1 + \alpha + \alpha^2)(1 + \beta + \beta^2) \geq 0$$

$$26. \text{ Let } f(x) = (x-3)^9 + (x-3^2)^9 + (x-3^3)^9 + \dots + (x-3^9)^9$$

$$\therefore f'(x) = 9 \left\{ (x-3)^8 + (x-3^2)^8 + (x-3^3)^8 + \dots + (x-3^9)^8 \right\} > 0$$

So, $f'(x) = 0$ has no real roots.

So, $f(x) = 0$ has one real and 8 imaginary roots

$$27. \text{ Roots} = \frac{4(2-i) \pm \sqrt{16(2-i)^2 + 8(1+i)(5+3i)}}{4(1+i)}$$

$$= -\frac{i}{1+i} \text{ or } \frac{4-i}{1+i} = \frac{-1-i}{2} \text{ or } \frac{3-5i}{2}$$

$$28. \quad xy = 2(x+y) \Rightarrow y(x-2) = 2x$$

$$\therefore y = \frac{2x}{x-2} \text{ but } x, y \in N$$

By trial, $x = 3, 4, 6$

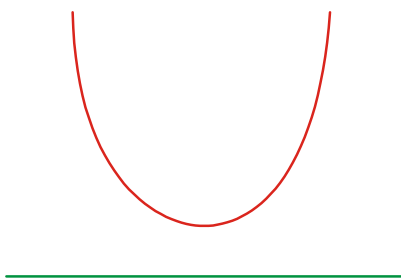
then $y = 6, 4, 3$ but $x \leq y$

$\therefore x = 3, 4$ and $y = 6, 4$ two solutions.

29. $\therefore$ Equation has no real roots

$$\therefore B^2 - 4AC < 0$$

$$\Rightarrow 16b^2 - 12ac < 0$$



$$\Rightarrow ac > \frac{4b^2}{3} > 0 \quad \therefore a > 0$$

$$\text{If } f(x) = 3ax^2 + 4bx + c \therefore f(x) > 0$$

$$\text{or } 3ax^2 + 4bx + c > 0, \forall x \in R$$

$$\therefore 3a(-1)^2 + 4b(-1) + c > 0$$

$$\Rightarrow 3a + c > 4b$$

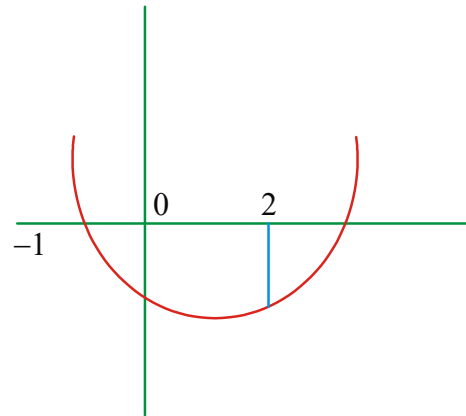
30. Let D_1 and D_2 be the discriminants of $ax^2 + bx + c = 0$ and $ax^2 - dx - c = 0$, respectively. Then,

$$D_1 = b^2 - 4ac \text{ and } D_2 = d^2 + 4ac$$

$$\text{Now, } D_1 + D_2 = b^2 + d^2 > 0$$

$\Rightarrow$ At least one of D_1 and D_2 is positive. Hence, at least one of the equation $ax^2 + bx + c = 0$ and $ax^2 - dx - c = 0$ has real roots. Thus, the given polynomial has at least two real roots.

31. Clearly, $f(-1) > 0, f(2) < 0$. Now,



$$f(0) = -4 < 0$$

$$\Rightarrow f(-1) = 1 - a - 4 > 0$$

$$\text{and } f(2) = 4 + 2a - 4 < 0$$

$$\Rightarrow a < -3 \text{ and } a < 0 \Rightarrow a \in (-\infty, -3)$$

$$32. (x-2y)^2 + (2y-3z)^2 + (3z-x)^2 \geq 0$$

$$33. 9x^2 + 6x + 1 \geq 0 \text{ and } 2-x > 0 \text{ and}$$

$$9x^2 + 6x + 1 < (2-x)^2$$

$$34. f\left(\frac{c}{2}\right) = c^2 - ac - c^2 + c + ac - b = c - b < 0$$

$$f\left(\frac{a}{2}\right) = a^2 - a^2 - ac + a + ac - b = a - b > 0$$

Hence a root lies between $\frac{c}{2}$ & $\frac{a}{2}$

$$35. \frac{2x}{2x^2 + 5x + 2} - \frac{1}{x+1} > 0$$

$$\Rightarrow \frac{3x+2}{(2x+1)(x+2)(x+1)} < 0$$

36. Eliminate 'z' and convert quadratic in y and use $\Delta \geq 0$

$$37. \text{ Let } 2^x = t; \Rightarrow t^2 + 2(a-1)t + a = 0$$

One root is positive $\Rightarrow x > 0 \Rightarrow t > 1$

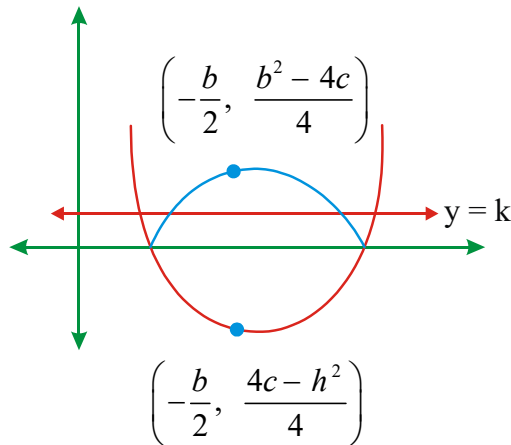
Another root is -ve $\Rightarrow x < 0 \Rightarrow t < 1$

$\therefore 1$ is a value between roots of t.

$$\therefore f(1) < 0 \Rightarrow a < \frac{1}{3}$$

38. $x^2 + bx + c = 0$ should have distinct roots $\Rightarrow \Delta > 0$

$$\Rightarrow b^2 - 4c > 0, k \in \left(0, \frac{b^2 - 4c}{4}\right)$$



$$39. a(x^2 - x + 1) - (x^2 + x + 1) \geq 0$$

$$\Rightarrow a \geq \frac{x^2 + x - 1}{x^2 - x + 1} \Rightarrow a \geq 1 + \frac{2}{x-1 + \frac{1}{x}}$$

But given $x \geq 2$

$$\Rightarrow a \geq 1 + \frac{2}{\left[\frac{3}{2}, \infty\right)} \Rightarrow a \in \left(1, \frac{7}{3}\right]$$

$$40. \alpha + \beta = -p; \alpha\beta = q$$

$$t = x^n \Rightarrow t^2 + p^n t + q = 0$$

Its roots are α^n, β^n ; $\therefore \alpha^n + \beta^n = -p^n$ (1)

Given $\frac{\alpha}{\beta}$ is a root of $x^n + 1 + (x+1)^n = 0$

$$\Rightarrow \frac{\alpha^n}{\beta^n} + 1 + \frac{(\alpha + \beta)^n}{\beta^n} = 0 \Rightarrow \alpha^n + \beta^n + (-p)^n = 0$$

$$\Rightarrow -p^n + (-p)^n = 0 \quad (\text{from (1)})$$

$\Rightarrow n$ must be even integer

$$41. x^2 + x + 1 = t \Rightarrow t = \left[\frac{3}{4}, \infty\right)$$

$$(t+1)^2 - (a-3)t(t+1) + (a-4)t^2 = 0$$

$$t^2[1-a+3+a-4] + t[2-a+3] + [1] = 0$$

$$\Rightarrow t(5-a) + 1 = 0 \Rightarrow 5-a = -\frac{1}{t}$$

$$\Rightarrow a = 5 + \frac{1}{t} = 5 + \left(0, \frac{4}{3}\right] = \left(5, \frac{19}{3}\right]$$

42. The equation $x^2 - px + r = 0$ has roots α, β and the equation $x^2 - qx + r = 0$ has roots $\frac{\alpha}{2}, 2\beta$

$$\Rightarrow r = \alpha\beta \text{ and } \alpha + \beta = p \text{ and } \frac{\alpha}{2} + 2\beta = q$$

$$\beta = \frac{2q-p}{3} \text{ and } \alpha = \frac{2(2p-q)}{3}$$

$$\Rightarrow \alpha\beta = r = \frac{2}{9}(2p-q)2p-q$$

43. Sum of roots $= \frac{\alpha^2 + \beta^2}{\alpha\beta}$ and product = 1

$$\text{given } \alpha + \beta = -p \text{ and } \alpha^3 + \beta^3 = q$$

$$\Rightarrow (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) = q$$

$$\therefore \alpha^2 + \beta^2 - \alpha\beta = \frac{-q}{p} \text{ and } (\alpha + \beta)^2 = p^2$$

$$\Rightarrow \alpha^2 + \beta^2 = \frac{p^3 - 2q}{3p}$$

$$\therefore \text{required equation is } x^2 - \frac{(p^3 - 2q)x}{(p^3 + q)} + 1 = 0$$

$$\Rightarrow (p^3 + q)x^2 - (p^3 - 2q)x + (p^3 + q) = 0$$

$$44. \frac{a_{10} - 2a_8}{2a_9} = \frac{(\alpha^{10} - \beta^{10}) - 2(\alpha^8 - \beta^8)}{2(\alpha^9 - \beta^9)}$$

$$= \frac{\alpha^8(\alpha^2 - 2) - \beta^8(\beta^2 - 2)}{2(\alpha^9 - \beta^9)}$$

$$= \frac{\alpha^8(6\alpha) - \beta^8(6\beta)}{2(\alpha^9 - \beta^9)} = \frac{6(\alpha^9 - \beta^9)}{2(\alpha^9 - \beta^9)} = 3$$

45. $x^2 + bx - 1 = 0$ and $x^2 + x + b = 0$ have a common root $\Rightarrow (1+b)^2 = (b^2 + 1)(1-b)$

$\Rightarrow b = 0, \pm\sqrt{3}i$

46. Let $f(x) = 2x^3 + 3x + k$

$f'(x) = 6x^2 + 3 > 0, \forall x \in \mathbb{R}$

$\Rightarrow f(x)$ is strictly increasing function.

$\Rightarrow f(x) = 0$ has only real root, so two roots are not possible.

47. Given equations are $x^2 + 2x + 3 = 0 \rightarrow (1)$

$ax^2 + bx + c = 0 \rightarrow (2)$ equations (1) and (2)

both have same roots, if $\frac{a}{1} = \frac{b}{2} = \frac{c}{3} \quad \therefore$

$a : b : c = 1 : 2 : 3$

48. Let $t = x - [x]$, then the given equations is

$-3t^2 + 2t + a^2 = 0 \Rightarrow t = \frac{1 \pm \sqrt{1 + 3a^2}}{3}$

$\therefore 0 \leq t \leq 1$, we have $0 \leq \frac{1 \pm \sqrt{1 + 3a^2}}{3} < 1$

Taking positive sign, we get $0 \leq \frac{1 + \sqrt{1 + 3a^2}}{3} < 1$

$\Rightarrow (a+1)(a-1) < 0 ; a \in (-1, 1)$

For no integral solution of a , we consider the interval $(-1, 0) \cup (0, 1)$

49. $\alpha + \beta = \frac{-q}{p}, \alpha\beta = \frac{r}{p} \dots (i)$

since, p, q and r are in AP $2q = p + r \dots (ii)$

$\frac{1}{\alpha} + \frac{1}{\beta} = 4 \Rightarrow \frac{\alpha + \beta}{\alpha\beta} = 4 \Rightarrow \alpha + \beta = 4\alpha\beta$

$\Rightarrow q = -4r$

On putting the value of q in in Eq(ii) we get

$p = -9r$ use $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

50. by simplyfying we get $\frac{(x-1)(x-6)}{(x+1)(x-4)} < 0$

Hence, $x \in (-1, 1) \cup (4, 6)$



LEVEL-IV

ASSERTION & REASON TYPE

1. Assertion (A): If $2x^2 + 3x + 4 = 0$ and $ax^2 + bx + c = 0$ have a common root, then $a : b : c = 2 : 3 : 4$

Reason(R): For a quadratic equation in x , complex roots occur in conjugate pairs.

- 1) Both A and R are true and R is the correct explanation of A.
2) Both A and R are true and R is not the correct explanation of A.
3) A is true but R is false
4) A is false but R is true

2. Assertion (A): The roots of

$(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$ are real.

Reason(R): Discriminant value above Quadratic equation is positive or equal to zero.

- 1) Both A and R are true and R is the correct explanation of A.
2) Both A and R are true and R is not the correct explanation of A.
3) A is true but R is false
4) A is false but R is true

3. (i) If α, β are the roots of $ax^2 + bx + c = 0$. If K is a real number [$K \neq 0, 1$] then the equation whose roots are $k\alpha, k\beta$ is $ax^2 + kbx + k^2c = 0$.

(ii) 8, 2 are roots of $x^2 + ax + \beta = 0$ and 3, 3 are the roots of $x^2 + \alpha x + b = 0$ then the roots of $x^2 + ax + b = 0$ are 9, 1.

Which of the above statement is correct

- 1) only (i) 2) only (ii)
3) Both (i) and (ii) 4) Neither (i) nor (ii)

4. Assertion(A): $x^2 + x + 1$ is greater than zero for all real x .

Reason (R): When $b^2 - 4ac < 0$. Then $a, ax^2 + bx + c$ have same sign for all real values of x .

- 1) Both A and R are true and R is the correct explanation of A.
2) Both A and R are true and R is not the correct explanation of A.
3) A is true but R is false
4) A is false but R is true

5. **Assertion (A):** The curve $y = x^2 - 6x + 9$ touches the x-axis.

Reason(R): $y = ax^2 + bx + c$ touches the x-axis if $\Delta = b^2 - 4ac = 0$.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

6. **Assertion (A)** If $x^2 - 7x + 10 < 0$ then $x \in (2, 5)$.

Reason (R): If 'a' and $ax^2 + bx + c$ have opposite signs and $b^2 - 4ac > 0$ then x lies between the roots of $ax^2 + bx + c = 0$.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

7. **Assertion (A):** The equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ has no root

Reason(R): $x-1 \neq 0$, then only above equation defined.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

8. **Assertion (A):**

$f(x) = ax^2 + bx + c$, for $a(\neq 0), b, c \in R$.
When $a > 0$ and $b^2 - 4ac < 0$, $f(x) = 0$ has complex roots.

Reason(R): The graph of $f(x) = ax^2 + bx + c$ cuts x-axis.

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

9. **Assertion (A):** If $f(x)$ is a quadratic expression such that $f(1) + f(2) = 0$, if -1 is a root of $f(x) = 0$ then the other root is $\frac{8}{5}$.

Reason (R): If $f(x) = ax^2 + bx + c$ then

$$\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$$

- 1) Both A and R are true and R is the correct explanation of A.
- 2) Both A and R are true and R is not the correct explanation of A.
- 3) A is true but R is false
- 4) A is false but R is true

10. **Assertion (A):** If $a > 0, b > 0$ and $c > 0$, then

$$(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$$

Reason (R): For positive numbers a, b, c, $AM \geq GM$

- 1) Both A and R are true and R is a consequence of A.
- 2) Both A and R are true and R is not a consequence of A.
- 3) A is true but R is false
- 4) Both A and R are false

11. **Assertion (A) :** If both roots of equation $4x^2 - 2x + a = 0, a \in R$, lie in the interval $(-1, 1)$, then $-2 < a \leq \frac{1}{4}$.

Reason (R) : If $f(x) = 4x^2 - 2x + a$, then

$$D \geq 0, f(-1) > 0 \text{ and } f(1) > 0 \Rightarrow -2 < a \leq \frac{1}{4}$$

- 1) Both A and R are true and R is a consequence of A.
- 2) Both A and R are true and R is not a consequence of A.
- 3) A is true but R is false
- 4) Both A and R are false

19. Observe the following lists:

List-I

List-II

A. $\sqrt{x^2 - 5x + 6} < -3 \Rightarrow x \in$ 1. $\left[-2, \frac{1}{2}\right]$

B. $2-3x-2x^2 \geq 0 \Rightarrow x \in$ 2. $(-1, 2)$

C. $x^2 - 3x + 2 > 0,$ 3. $[-1, 2)$

$x^2 - 3x - 4 \leq 0 \Rightarrow x \in$

D. $\frac{x+1}{x-2} < 0 \Rightarrow x \in$ 4. ϕ

5. $[-1, 1) \cup (2, 4]$

The correct match for List-I from List-II is

A B C D

A B C D

1) 4 1 5 2

2) 4 5 1 2

3) 1 2 3 4

4) 2 3 4 5

20. Let $x^2 - (m-3)x + m = 0, (m \in R)$ be a quadratic equation. Match the following.

List - I

List - II

1) Both roots are greater than 2 A) $m \in (10, \infty)$

2) Both roots are smaller than 2 B) $m \in \phi$

3) Exactly one root lies in $(1, 2)$ C) $m \in [9, 10)$

4) Both roots lie in the interval $(1, 2)$ D) $m \in (-\infty, 1]$

I

II

III

IV

1) C

D

A

B

2) D

C

A

B

3) C

D

B

A

4) C

B

D

A

INTEGER ANSWER TYPE QUESTIONS

21. The number of solutions of the equation

$\sqrt{x+1} - \sqrt{x-1} = \sqrt{4x-1}$ is _____

22. The number of solution of

$2^{\sin(|x|)} = 4^{|\cos x|}$ in $[-\pi, \pi]$ is equal to

LEVEL - IV - KEY

1) 1	2) 1	3) 3	4) 1
5) 1	6) 1	7) 1	8) 3
9) 2	10) 1	11) 1	12) 3
13) 1	14) 1	15) 4	16) 4
17) 2	18) 4	19) 1	20) 1
21) 0	22) 4		

LEVEL - IV - HINTS

1. for $2x^2 + 3x + 4 = 0$

$\Delta = 9 - 16 < 0$

roots are complex conjugate

$\Rightarrow ax^2 + bx + c = 0$, and

$2x^2 + 3x + 4 = 0$

have both roots common.

2. $\Delta = 2[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$

3. (i) $f\left(\frac{x}{k}\right) = 0$

(ii) $-a = 10, \beta = 16,$

$-\alpha = 6, b = 9$

$\therefore x^2 + ax + b = x^2 - 10x + 9 = 0$

$x = 1, 9$

4. $\Delta < 0$, a & $ax^2 + bx + c$ have opposite signs

5. $\Delta = 0 \Rightarrow y = f(x)$ touches x-axis

6. $(x-2)(x-5) < 0$

7. $x = 1$ not satisfies.

8. from the synopsis.

9. $f(1) + f(2) = 0$

$\Rightarrow 5a + 3b + 2c = 0$

-1 is a root $\Rightarrow 3a - 3b + 3c = 0$

$\Rightarrow 8a + 5c = 0$

$\Rightarrow \frac{c}{a} = \frac{-8}{5}$

10. $AM \geq GM$

$$\Rightarrow \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} \geq \left(\frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c} \right)^{\frac{1}{3}} \rightarrow (1)$$

$$\frac{(a+b+c)}{3} \geq (abc)^{\frac{1}{3}} \rightarrow (2)$$

multiply (1) and (2)

11. $A : \Delta > 0$, and $f(-1) > 0$ and $f(1) > 0$

12. (i) let $f(x) = x^2$

then $g(x) = x^2 + 2x + 2$

(ii) $\frac{p}{2} = \frac{-7}{q} = \frac{3p}{6}$

13. $A = x^2 - 5x + 6 = (x-2)(x-3) = 0 \Rightarrow x = 2, 3$

$B = |x|^2 - 7|x| = |x|(|x| - 7) = 0 \Rightarrow x = 0, \pm 7$

$C = \Delta = 16 - 20 < 0$, no real root

$D = |x|^2 - 5|x| + 6 = (|x| - 2)(|x| - 3) = 0$
 $\Rightarrow x = \pm 2, \pm 3$

14. for $ax^2 + bx + c = 0$, $\alpha\beta = \frac{c}{a}$.

15. If the roots of a quadratic equation $ax^2 + bx + c = 0$ are in ratio $m:n$, then

$$\frac{(m+n)^2}{mn} = \frac{b^2}{ac}$$

16. $A : \alpha + \beta = -\frac{b}{a} < 0, \alpha\beta = \frac{c}{a} > 0$

$B : \alpha\beta = \frac{c}{a} = 1$

$C : a + b + c = 0 \Rightarrow a(1)^2 + b(1) + c = 0$

$D : a + c = b \Rightarrow a(-1)^2 + b(-1) + c = 0$

17. $A : (x-3)(x-1) > 0 \Rightarrow x < 1$ or $x > 3$

$B : (x-2)(x-3) \leq 0 \Rightarrow 2 \leq x \leq 3$

$C : x^2 + 6x - 27 > 0 \Rightarrow (x+9)(4-3) > 0$

$\Rightarrow x < -9$ or $x > 3$

$-x^2 + 3x + 4 > 0 \Rightarrow (x+1)(x-4) < 0 \Rightarrow -1 < x < 4$

$\therefore 3 < x < 4$

$D : x^2 - 3x - 4 < 0 \Rightarrow (x+1)(x-4) < 0 \Rightarrow -1 < x < 4$

$x^2 - 3x + 2 > 0 \Rightarrow (x-2)(x-1) > 0 \Rightarrow x < 1$ or $x > 2$

$\therefore x \in (-1, 1) \cup (2, 4)$

18. If $\alpha : \beta = m : n$ then $\frac{(m+n)^2}{mn} = \frac{b^2}{ac}$

(i) $m : n = 1 : 1$

(ii) $m : n = 2 : 1$

(iii) $m : n = 3 : 1$

(iv) $\alpha = \beta^n \Rightarrow (ac^2)^{\frac{1}{n+1}} + (a^2c)^{\frac{1}{n+1}} + b = 0$

19. (i) $L.H.S > 0$ but $R.H.S < 0$

(ii) $(x+2)(2x-1) \leq 0 \Rightarrow x \in \left[-2, \frac{1}{2}\right]$

(iii) $x^2 - 3x + 2 > 0 \Rightarrow (x-2)(x-1) > 0 \Rightarrow x < 1$ or $x > 2$

$x^2 - 3x - 4 \leq 0 \Rightarrow (x-4)(x+1) \leq 0 \Rightarrow -1 \leq x \leq 4$

(iv) $(x+1)(x-2) < 0$ and $x - 2 \neq 0$

20. I) $f(2) > 0, D \geq 0, \alpha + \beta > 4$

II) $f(2) > 0, D \geq 0, \alpha + \beta < 4$

III) $f(1) \cdot f(2) < 0, D \geq 0$

IV) $f(1) > 0, f(2) > 0, D \geq 0, 2 < \alpha + \beta < 4$

21. S.O.B.S

22. $\sin|x| = 2|\cos x|$

$\Rightarrow \sin x = 2\cos x$ in Q_1

$\sin x = -2\cos x$ in Q_2

